

# Yalla: Estimating Muscle Growth, Maintenance, and Decline from Training Logs

Oliver Frings

Version 2.1 (v121) 17 July 2026

## Abstract

Yalla estimates, per muscle and for the body overall, whether recent training is enough to grow, hold, or slowly lose muscle size. The estimate reads *training stimulus* from logged sets, loads, and reps; the app never sees the body itself. Two axes drive it: a *dose* axis (effort-weighted weekly sets against volume landmarks) and a *trend* axis (progressive overload measured as the better of a set trend and a load trend). Set effort is captured with a three-level control whose default is estimated from the load–rep relationship against the lifter’s own best set. This document defines every quantity, the equations that combine them, the decision rules that produce a state, and the evidence each parameter rests on. It also states the model’s assumptions, its validation status, and where it is deliberately conservative. The specification corresponds to the functions `growthStatus()`, `inferEffort()`, and `planForecast()` in `app.js`.

## 1 Introduction

Most training apps record what was lifted; few say what it means for muscle. Yalla estimates, per muscle and for the body overall, whether recent training is enough to grow, maintain, or slowly lose size, using only the sets, loads, and reps a user logs. This document specifies that model: the quantities it computes, the equations that combine them, the decision rules that produce a per-muscle verdict, and the evidence behind each parameter. The goal is an auditable feedback signal whose parameters trace to the resistance-training literature. Sections 3–11 define the model, Section 13 lists the evidence with a grade for each source, and Section 14 describes the checks that accompany the model in the absence of a body-composition study.

## 2 Background and related work

The model synthesises several strands of resistance-training research.

*Volume and its counting.* Schoenfeld, Ogborn & Krieger [13] established a graded dose–response between weekly set volume and hypertrophy, with each added set contributing on average about +0.37% of muscle across the studied range. The larger meta-regression of Pelland et al. [9] (67 studies, 2,058 subjects) confirmed rising-but-diminishing returns, found that counting indirect (secondary) sets at about half best fit the data, and found frequency largely neutral once weekly volume is matched. Yalla adopts fractional set counting (1) and derives its volume landmarks from a continuous dose–response curve (9) built on these findings.

*Effort and proximity to failure.* Sets far from failure contribute little; Refalo et al. [11] report that hypertrophy improves as sets approach failure, with roughly comparable outcomes across zero to four reps in reserve. Yalla scales each set by an effort factor (2) and, because reps in reserve cannot be read from load and reps alone, estimates it by default from the load–rep relationship [3] against the lifter’s own best set.

*Landmarks, maintenance, and decline.* The MEV/MAV volume-landmark framework popularised by Israetel et al. [5] informs the landmark structure, though as a practitioner synthesis it carries the weakest evidence grade here. Bickel et al. [1] showed that built muscle is retained on a small fraction of the volume that built it, which sets the maintenance floor; Mujika & Padilla [6] documented the timeline over which an insufficient stimulus leads to loss, which motivates treating a sustained low dose as decline. Plotkin et al. [10] found that progressing load or reps drives near-identical growth, which is why the trend axis credits progress from either.

*Positioning.* These landmarks and principles are widely used as coaching guidance but are rarely turned into a per-log computed signal. Yalla operationalises them into a transparent, per-muscle estimate with stated uncertainty and a reproducible verification suite (Section 14).

### 3 Model overview

The unit of analysis is a muscle group  $g$  (chest, lats, quads, and so on). Training history is a set of logged entries; each entry  $e$  records one exercise in one session with a date  $d_e$ , a top-set load  $w_e$  (kg), top-set reps  $r_e$ , a set count  $n_e$ , an effort code  $f_e \in \{0, 1, 2\}$ , and a stored volume-load  $v_e$ .

Each muscle is placed in one of three states, combined from the two axes: *growing* (adequate dose and progressing), *holding* (a maintenance dose, or flat), and *under-stimulated* (below the volume needed to hold size). The dose axis answers whether there is enough stimulus this week; the trend axis answers whether the stimulus is rising, flat, or falling. A muscle grows only when both are satisfied: enough sets and a rising trend. This mirrors the requirement that hypertrophy needs both sufficient volume and progressive overload.

## 4 Volume attribution

### 4.1 Fractional set counting

Each exercise contributes to its primary muscle at full weight and to each secondary muscle at half. Writing the attribution weight of exercise  $x$  to muscle  $g$  as  $\alpha(x, g)$ :

$$\alpha(x, g) = \begin{cases} 1 & \text{if } g \text{ is the primary mover of } x, \\ 0.5 & \text{if } g \text{ is a secondary mover of } x, \\ 0 & \text{otherwise.} \end{cases} \quad (1)$$

The 0.5 weight for indirect work is the fractional counting scheme that best fit the hypertrophy data in the Pelland et al. dose-response meta-regression [9].

### 4.2 Effort weighting

Only sets taken reasonably close to failure drive much growth, so each set is scaled by an effort factor  $\varepsilon(f)$  before it counts:

$$\varepsilon(f) = \begin{cases} 0.5 & \text{Easy } (f=0, \geq 4 \text{ reps in reserve}), \\ 1 & \text{Hard } (f=1, \sim 1\text{--}3 \text{ in reserve; the default}), \\ 1 & \text{Max } (f=2, 0\text{--}1 \text{ in reserve}). \end{cases} \quad (2)$$

Max carries no growth premium over Hard: hypertrophy is comparable across roughly 0–4 reps in reserve, and training to failure mainly adds fatigue [11]. Max still helps the load trend indirectly through the extra reps it buys. Entries with no recorded effort (all logs predating the feature) default to Hard, so  $\varepsilon = 1$  and historical counts are unchanged.

### 4.3 Weekly effective sets per muscle

Time is bucketed into  $N = 10$  rolling weeks; week index  $k = 9$  is the current week. The effective weekly sets for muscle  $g$  in week  $k$  sum every contributing entry:

$$S_{g,k} = \sum_{e \in \text{week } k} n_e \varepsilon(f_e) \alpha(x_e, g). \quad (3)$$

This same effort-weighted, fractional count feeds the weekly-sets chart, the balance radar, and the “hard sets” ring, so every surface in the app reads consistently.

## 5 Effort inference

Proximity to failure cannot be recovered from load and reps alone: the same  $100 \text{ kg} \times 8$  can be an all-out set or an easy one. So effort is a self-report. To keep friction to one optional tap, its default is estimated from the numbers and shown pre-selected; the lifter overrides only when the estimate is wrong.

### 5.1 Estimated one-rep max (anchor)

Set strength is summarised by the Epley one-rep-max estimate [3], with a rep denominator  $K$  (population default  $K = 30$ ; Section 5.3 calibrates it per lifter):

$$\varphi(w, r) = w \left( 1 + \frac{r}{K} \right). \quad (4)$$

The anchor for exercise  $x$  is the best estimate over the lifter’s *prior* sessions for that exercise (the current, in-progress session is excluded):

$$A_x = \max_{e \in \text{hist}(x)} \varphi(w_e, r_e). \quad (5)$$

### 5.2 Reps in reserve and classification

Inverting (4) gives the reps a lifter should complete at load  $w$  before failure; subtracting the reps actually done estimates reps in reserve:

$$\text{RIR} = K \left( \frac{A_x}{w} - 1 \right) - r, \quad (6)$$

evaluated on the session’s hardest set for that exercise (the one with the largest  $\varphi$ ). The estimate maps to an effort code:

$$f = \begin{cases} 2 \text{ (Max)} & \text{if } \text{RIR} \leq 1, \\ 0 \text{ (Easy)} & \text{if } \text{RIR} \geq 4, \\ 1 \text{ (Hard)} & \text{otherwise.} \end{cases} \quad (7)$$

The anchor is recomputed on every estimate. Because it rolls, it adapts to deloads and off days and cannot lock in a wrong baseline, the failure mode of a one-time calibration. When there is no prior baseline for the lift, or the exercise carries no external load (bodyweight or timed holds), (6) is not evaluated and the default stays Hard.

### 5.3 Self-calibrated rep denominator

The denominator  $K$  in (4) varies by lifter and by lift. A set logged as Max is a near-true failure point, so two Max sets of the same exercise at different loads pin  $K$ : setting  $\varphi$  equal for both,  $w_1(1 + r_1/K) = w_2(1 + r_2/K)$ , which solves to

$$K = \frac{w_2 r_2 - w_1 r_1}{w_1 - w_2}. \quad (8)$$

Valid within-exercise pairs are pooled across the whole log, the median is taken and clamped to  $[20, 40]$ , and  $K = 30$  is used until enough failure data exists. This grounds the effort estimate in the lifter’s own data.

## 6 Volume landmarks

Four thresholds, in effective sets per muscle per week, define the dose axis. They follow the MEV/MAV framework and the dose–response meta-analyses.

| Symbol (value) | Name                     | Meaning                                                                                       |
|----------------|--------------------------|-----------------------------------------------------------------------------------------------|
| MV (2)         | Maintenance volume       | Floor to hold existing size. Below it, a previously-trained muscle reads as under-stimulated. |
| MEV (6)        | Minimum effective volume | Lower edge of the productive growth window; at or above it, an adequate dose is present.      |
| target (10)    | Practical target         | Where most of the dose–response benefit has accrued; the balance-radar goal.                  |
| MAV (20)       | Upper productive bound   | Beyond it, added sets mostly add fatigue with little further growth.                          |

A single set of landmarks is applied to every muscle. Real per-muscle landmarks differ, but one baseline keeps the output legible. The maintenance floor is deliberately low: built muscle is held on very little (as little as roughly one-ninth of the building volume in young trained adults [1]), so a 2-set floor avoids flagging loss too early.

### 6.1 Continuous backbone

The four landmarks discretise a continuous relationship [13, 9]. We model the fraction of the attainable per-muscle growth stimulus at  $s$  effective sets per week as a saturating curve  $\rho(s)$ , calibrated so the practical target captures about 80% of what is attainable:

$$\rho(s) = 1 - e^{-s/s_0}, \quad s_0 = \frac{\text{target}}{\ln 5} \approx 6.21. \quad (9)$$

The landmarks are read off this single curve:  $\rho(\text{MV}) \approx 0.28$ ,  $\rho(\text{MEV}) \approx 0.62$ ,  $\rho(\text{target}) \approx 0.80$ , and  $\rho(\text{MAV}) \approx 0.96$ . So MEV is the  $\approx 0.6$  knee, target the 0.8 point, and MAV the  $\approx 0.95$  near-plateau. This replaces four independent thresholds with one meta-analytic function; the growth signal also reports  $\rho(D_g)$  as a per-muscle share of the attainable growth stimulus.

## 7 Temporal windows

Both axes compare a recent window against an earlier one, each a three-week mean over the 10-week series. For any weekly series  $X_g$ :

$$\text{recent}(X_g) = \frac{1}{3} \sum_{k=7}^9 X_{g,k}, \quad \text{earlier}(X_g) = \frac{1}{3} \sum_{k=4}^6 X_{g,k}. \quad (10)$$

The three-week mean smooths single missed or heavy sessions. The recent window is weeks 8–10 (the last three), the earlier window is weeks 5–7; the oldest three weeks provide history for charts but do not enter the comparison. Because the window is recent, the signal responds within about three weeks of a genuine change in training.

## 8 The two axes

### 8.1 Dose

The current dose for muscle  $g$  is the recent mean of effective weekly sets:

$$D_g = \text{recent}(S_g). \quad (11)$$

### 8.2 Trend: progressive overload two ways

Overload can come from more effective sets or from heavier load; either counts as progress [10, 12]. A set trend and a load trend are formed as recent-over-earlier ratios. The load series  $L_{g,k}$  is the best estimated one-rep max reached in week  $k$  on exercises for which  $g$  is the primary mover:

$$T_g^S = \frac{\text{recent}(S_g)}{\text{earlier}(S_g)}, \quad T_g^L = \frac{\text{recent}(L_g)}{\text{earlier}(L_g)}. \quad (12)$$

Each ratio is defined only when its earlier window is positive; otherwise it is treated as absent. The combined trend takes the better of the two:

$$\tau_g = \max(T_g^S, T_g^L), \quad (13)$$

with an absent ratio contributing 0 to the maximum, and  $\tau_g$  itself absent only when both ratios are absent (a muscle just started, with no earlier baseline). Taking the maximum prevents classic strength progression (add load, drop reps) from being misread as a decline: raw tonnage would fall there while  $T^L$  rises.

An earlier version used tonnage,  $V = w r n$ , as the sole trend signal. Tonnage rewards total kilograms moved, so lighter high-rep work inflates it while heavier low-rep work deflates it; it is a poor proxy for stimulus, and (13) replaces it.

## 9 Per-muscle state

The state  $\sigma_g$  is a function of the dose  $D_g$  and the trend  $\tau_g$ , evaluated top to bottom (first matching rule wins):

1. **Under-stimulated** if  $D_g < \text{MV}$  (below the maintenance floor of 2).
2. **Under-stimulated** if  $\tau_g$  exists and  $\tau_g < 0.75$  (training eased off sharply).
3. Otherwise, if  $D_g \geq \text{MEV}$  (adequate dose,  $\geq 6$ ): **growing** if  $\tau_g$  is absent or  $\tau_g \geq 1.08$ ; **holding** otherwise ( $0.75 \leq \tau_g < 1.08$ ).
4. Otherwise (maintenance band,  $2 \leq D_g < 6$ ): **growing** if  $\tau_g \geq 1.15$  and  $D_g \geq 5$  (climbing toward a growth dose); **holding** otherwise.

The asymmetry is intentional: a muscle grows only with both an adequate dose and a rising trend, but it can be flagged under-stimulated by a low dose alone. Holding is the wide middle band: enough volume to keep size, below the dose that adds it.

## 9.1 Borderline states

The inputs carry known noise: effort-weighted sets to about  $\pm 1$  set, and each trend ratio to about  $\pm 0.06$ . When a muscle’s dose or trend sits within that range of a decision boundary, it is flagged *borderline* and shown with a tilde to mark the verdict as provisional. The sensitivity analysis (Section 14) shows the growth-trend boundary is where this matters most.

## 10 Overall verdict

Let  $n_{\text{grow}}$ ,  $n_{\text{hold}}$ ,  $n_{\text{shrink}}$  count trained muscles in each state, and  $m$  be the number of trained muscles. Let  $\tau_{\text{all}}$  be the recent-over-earlier ratio of whole-body weekly volume. The headline is:

1. **Need more data** if fewer than 3 weeks show any training, or  $m = 0$ .
2. **At risk** if  $n_{\text{shrink}} > n_{\text{grow}}$  and  $n_{\text{shrink}} \geq \lceil m/3 \rceil$ .
3. **Growing** if  $n_{\text{grow}} \geq \max(1, \text{round}(0.4m))$  and  $\tau_{\text{all}}$  is absent or  $\geq 0.95$ .
4. **Holding** otherwise.

The “at risk” headline requires under-stimulated muscles to both outnumber growing ones and reach a third of those trained, so a few neglected small muscles do not flip the whole read. It resolves to “at risk” chiefly when most of recent training is genuinely sparse.

## 11 Plan forecast

Separately from the history-based signal, the plan forecast projects a plan’s *prescribed* weekly sets before any of it is logged. Prescribed sets per rotation are scaled to a weekly rate by how often the plan is run, capped so an ambitious frequency on a short rotation cannot over-promise:

$$\hat{s}_g = s_g \cdot \min\left(1.2, \frac{p}{W}\right), \quad (14)$$

where  $s_g$  is prescribed sets per rotation (primary plus half of secondary, as in (1)),  $p$  is planned sessions per week, and  $W$  is the number of distinct rotation workouts. Each trained muscle is then sorted into under-dosed ( $< MV$ ), growth window (MEV to 20), past-productive ( $> 20$ ), or maintenance, and the counts drive a plain-language outlook. The forecast promises no centimetres or kilograms; it reports the dose against the growth window and what to change.

## 12 Simulations

The model’s behaviour is clearest as projected muscle gain under different training regimes. A seeded, reproducible simulation (`research/simulate.mjs`) drives the Monte Carlo forecast below forward for each regime, in the app’s colours; every number regenerates from the same seed.

### 12.1 Monte Carlo forecast

A point prediction of muscle gain would overstate what the model knows, so the in-app forecast is probabilistic. Each of 500 runs accumulates a weekly fractional gain

$$\Delta_t = b \cdot m \cdot \alpha \cdot \rho(D) \cdot \eta \cdot o \cdot e^{-t/32}, \quad (15)$$

where  $\rho(D)$  is the dose–response of Section 4 [13, 9],  $\eta$  scales with proximity to failure [11],  $o$  is near 1 while progressing and drops without added load [10],  $b$  is a training-age base rate [2],

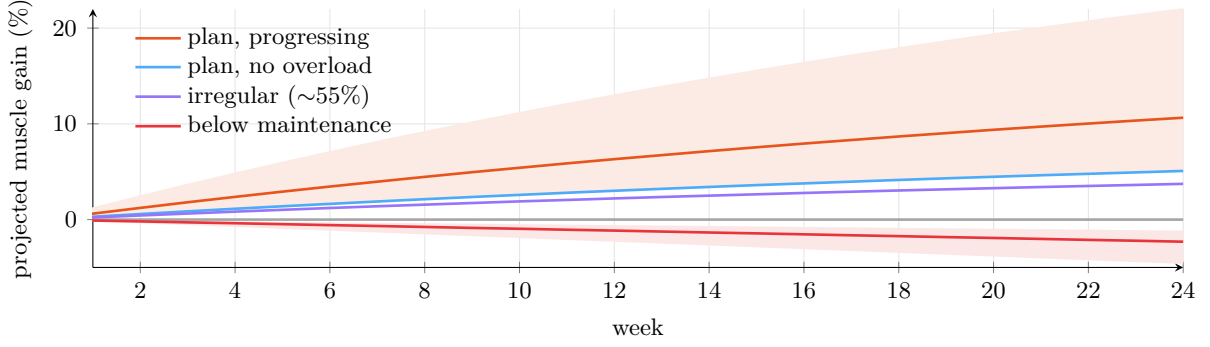


Figure 1: Projected muscle gain over 24 weeks under four regimes at a novice base rate (median lines; shaded 10–90% bands on the best and worst). Same volume drives the progressing and no-overload lines, so their gap is progressive overload; below maintenance the median crosses zero (net loss). Bands are chiefly inter-individual variability [4].

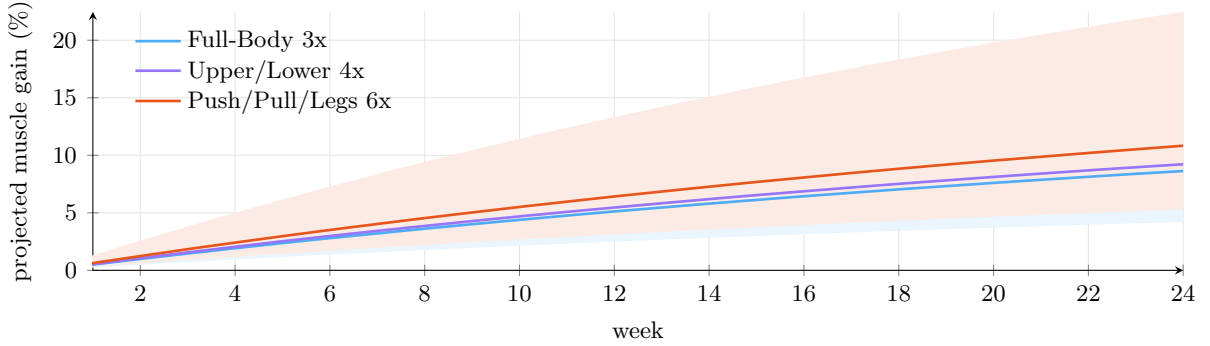


Figure 2: Projected muscle gain for three standard plans, each executed with progressive overload (median, 10–90% bands). More weekly volume per muscle projects more gain, with diminishing returns and heavily overlapping bands.

$\alpha$  is an age factor that attenuates hypertrophy past about 30 [7], and  $m$  is the inter-individual response multiplier, drawn lognormal to span the range Hubal et al. [4] observed (cross-sectional-area change from about  $-2\%$  to  $+59\%$ ). Sex enters as a separate lever but is set near neutral: relative hypertrophy is comparable between sexes [8], so it informs absolute expectations rather than the percentage gain shown here. Below maintenance the increment turns negative (atrophy [6]), so a starved scenario trends down and the axis extends below zero. The first terms are well constrained by the literature;  $b$  is literature-informed with a wide prior, and  $m$  is the dominant source of spread, so the bands, not the median, carry the message. Figures 1 and 2 apply this forecast across regimes and plans; the same model drives the plan-versus-pace forecast in the app.

Beyond the individual-response spread, Figure 3 shows how the median 16-week gain moves when each input is swung across a plausible range (the app draws the same tornado). Training experience and progression move it most; volume and age move it least over the ranges tested. This is a forecast-level companion to the threshold sensitivity of Section 14.

### 13 Scientific basis

Each parameter and mechanism traces to a specific source. Effect sizes are context-dependent; the model uses the literature to set its structure and thresholds, and promises no outcomes. Grades are GRADE-style ratings of the underlying evidence: High (meta-analyses of RCTs),

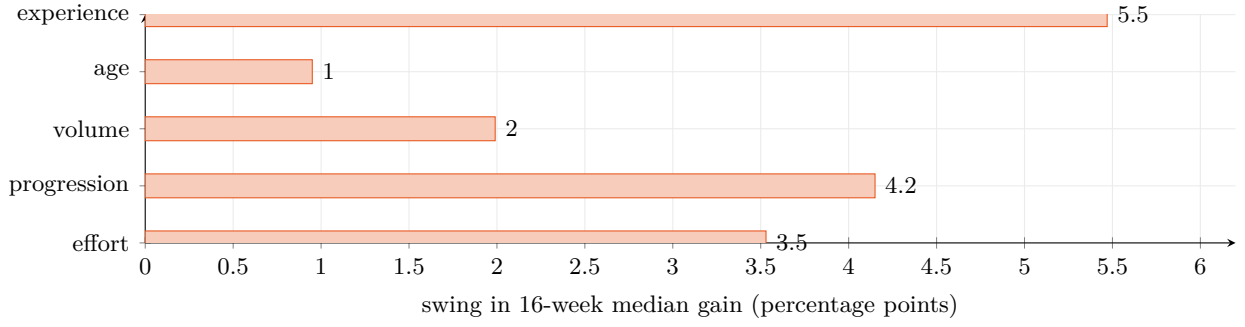


Figure 3: Parameter sensitivity of the Monte Carlo forecast: how far the median 16-week gain swings as each input moves across a plausible range, around a novice plan baseline.

Moderate (individual RCTs or focused reviews), Low (practitioner framework). A Low grade flags a parameter that rests on weaker evidence and is the first candidate for revision.

| Mechanism / parameter                                                                          |                                                                          |       | Basis                                                                                                                       | Grade    |
|------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------|-------|-----------------------------------------------------------------------------------------------------------------------------|----------|
| Graded response;                                                                               | volume ~10 sets as a practical target                                    | dose- | Schoenfeld, Ogborn & Krieger [13]: each added weekly set $\approx +0.37\%$ muscle, benefit rising across the studied range. | High     |
| Fractional (0.5) indirect sets; diminishing returns; frequency neutral at equal volume         | counting of sets; diminishing returns; frequency neutral at equal volume |       | Pelland et al. [9] (peer-reviewed; 67 studies, 2,058 subjects).                                                             | High     |
| Continuous backbone $\rho(s)$ and landmark derivation                                          |                                                                          |       | Derived from the dose-response curves above [13, 9].                                                                        | High     |
| Effort factor $\varepsilon$ ; hypertrophy comparable across ~0–4 RIR, dropping off when easier |                                                                          |       | Refalo et al. [11], systematic review and meta-analysis of proximity-to-failure.                                            | High     |
| MEV/MAV landmark structure (~10 to 20 sets)                                                    |                                                                          |       | Israetel et al. [5], volume-landmarks framework.                                                                            | Low      |
| Maintenance floor MV; size held on ~1/3 (older) to ~1/9 (young) of building volume             |                                                                          |       | Bickel, Cross & Bamman [1] (RCT, 16-wk build plus 32-wk reduced dose).                                                      | Moderate |
| Decline below maintenance stimulus; ~2–4-week strength retention, then loss                    |                                                                          |       | Mujika & Padilla [6] detraining review; cross-sectional-area loss from ~2–3 weeks of cessation.                             | Moderate |
| Overload via added load <i>or</i> reps; the trend tracks both                                  |                                                                          |       | Plotkin et al. [10] (RCT; load- and rep-progression gave near-identical growth); mechanism in Schoenfeld [12].              | Moderate |
| Epley one-rep-max $\varphi$ ; self-calibrated denominator $K$                                  |                                                                          |       | Standard load-rep relationship [3]; $K$ fitted from the lifter’s own Max sets (8).                                          | Method   |
| Monte Carlo forecast: training-age base rate                                                   |                                                                          |       | Damas et al. [2]; hypertrophy rate depends on training status.                                                              | Moderate |
| Monte Carlo forecast: inter-individual response spread                                         |                                                                          |       | Hubal et al. [4] (n=585); 12-wk CSA change $\approx -2\%$ to $+59\%$ .                                                      | High     |
| Forecast age factor                                                                            |                                                                          |       | Peterson et al. [7]; lean-mass gains attenuate with age.                                                                    | Moderate |
| Forecast sex lever ( $\approx$ neutral for % gain)                                             |                                                                          |       | Roberts et al. [8]; relative hypertrophy comparable between sexes.                                                          | High     |

## 14 Verification

No body-composition study validates the model (see the limitations that follow). Three reproducible checks accompany it (the **research/** folder in the repository), each raising rigor without



new measurement.

- **Known-answer tests.** Canonical scenarios — a returning lifter, a deload, easy versus hard sets, a hard stop — paired with domain-expected verdicts, so miscalibration or later drift is caught in code.
- **Sensitivity analysis.** Perturbing each threshold by  $\pm 10\%$  over a synthetic population shows the growth-trend cutoff is the most load-bearing parameter (about a quarter of muscles change state), while the maintenance floor and sharp-drop cutoff are nearly cosmetic. This is why the borderline band sits on the trend boundary.
- **Predictive-coherence backtest.** On a training log, muscles labelled growing should precede larger subsequent gains in estimated 1RM than those labelled holding or under-stimulated. Strength change is only a proxy for hypertrophy; a body-composition study remains the standard these checks cannot replace.

## 15 Assumptions and limitations

- **Validation status.** This is a mechanistic model, grounded in the literature and internally consistent. It has no validation against direct body-composition measurements (DXA, ultrasound, or MRI): its thresholds are derived from published dose-response findings, none are fitted to an outcome dataset, and it claims no predictive-accuracy interval. The checks in Section 14 cover internal consistency and coherence; external accuracy stays out of scope.
- **A stimulus estimate.** Every output estimates the training stimulus from logs. The app cannot see the body; no measurement of muscle mass is implied.
- **Effort is self-reported.** Reps in reserve is not recoverable from load and reps alone; (6) sets only a default. Day-to-day max-rep variability ( $\pm 1\text{--}3$  reps) sits at the resolution of the effort bands, so the estimate is noisy near failure and the manual override remains the ground truth.
- **Top-set inference.** History stores each exercise’s top set and its set count, without the reps of every set, so within-session rep drop-off is not yet used as an effort cue.
- **One landmark set for all muscles.** Chosen for legibility; real landmarks vary by muscle and training age.
- **Load trend needs external load.** Bodyweight and timed work do not contribute to  $T^L$ ; those muscles rely on the set trend alone.
- **Residual tonnage use.** The overall verdict’s  $\tau_{\text{all}}$  still uses whole-body volume; the per-muscle trend does not.
- **Conservative by design.** The maintenance floor and the “at risk” threshold are set conservatively, biasing the model toward under-reporting muscle loss.

## References

- [1] Bickel CS, Cross JM, Bamman MM. Exercise dosing to retain resistance-training adaptations in young and older adults. *Med Sci Sports Exerc.* 2011;43(7):1177–1187. doi:10.1249/MSS.0b013e318207c15d
- [2] Damas F, Phillips SM, Libardi CA, et al. Resistance training-induced changes in integrated myofibrillar protein synthesis are related to hypertrophy only after attenuation of muscle damage. *J Physiol.* 2016;594(18):5209–5222. doi:10.1113/JP272472

- [3] Epley B. Poundage chart. *Boyd Epley Workout*. Lincoln, NE: Body Enterprises; 1985.
- [4] Hubal MJ, Gordish-Dressman H, Thompson PD, et al. Variability in muscle size and strength gain after unilateral resistance training. *Med Sci Sports Exerc.* 2005;37(6):964–972. doi:10.1249/01.mss.0000170469.90461.5f
- [5] Israetel M, Hoffmann J, Smith CW. Training volume landmarks for muscle growth (MEV/MAV/MRV). *Renaissance Periodization*; 2017. <https://rpstrength.com/blogs/articles/training-volume-landmarks-muscle-growth>
- [6] Mujika I, Padilla S. Detraining: loss of training-induced physiological and performance adaptations. Part I — short-term insufficient training stimulus. *Sports Med.* 2000;30(2):79–87. doi:10.2165/00007256-200030020-00002
- [7] Peterson MD, Sen A, Gordon PM. Influence of resistance exercise on lean body mass in aging adults: a meta-analysis. *Med Sci Sports Exerc.* 2011;43(2):249–258. doi:10.1249/MSS.0b013e3181eb6265
- [8] Roberts BM, Nuckols G, Krieger JW. Sex differences in resistance training: a systematic review and meta-analysis. *J Strength Cond Res.* 2020;34(5):1448–1460. doi:10.1519/JSC.0000000000003521
- [9] Pelland JC, Remmert JF, Robinson ZP, Hinson SR, Zourdos MC. The resistance training dose response: meta-regressions exploring the effects of weekly volume and frequency on muscle hypertrophy and strength gains. *Sports Med.* 2026. doi:10.1007/s40279-025-02344-w
- [10] Plotkin D, Coleman M, Van Every D, et al. Progressive overload without progressing load? The effects of load or repetition progression on muscular adaptations. *PeerJ.* 2022;10:e14142. doi:10.7717/peerj.14142
- [11] Refalo MC, Helms ER, Trexler ET, Hamilton DL, Fyfe JJ. Influence of resistance training proximity-to-failure on skeletal muscle hypertrophy: a systematic review with meta-analysis. *Sports Med.* 2023;53(3):649–665. doi:10.1007/s40279-022-01784-y
- [12] Schoenfeld BJ. The mechanisms of muscle hypertrophy and their application to resistance training. *J Strength Cond Res.* 2010;24(10):2857–2872. doi:10.1519/JSC.0b013e3181e840f3
- [13] Schoenfeld BJ, Ogborn D, Krieger JW. Dose-response relationship between weekly resistance training volume and increases in muscle mass: a systematic review and meta-analysis. *J Sports Sci.* 2017;35(11):1073–1082. doi:10.1080/02640414.2016.1210197

## A Symbol reference

| Symbol               | Definition                                                                                      |
|----------------------|-------------------------------------------------------------------------------------------------|
| $g, x, e$            | Muscle group, exercise, logged entry                                                            |
| $w, r, n, f$         | Top-set load (kg), top-set reps, set count, effort code $\{0, 1, 2\}$                           |
| $\alpha(x, g)$       | Muscle attribution weight: 1 primary, 0.5 secondary, 0 otherwise (1)                            |
| $\varepsilon(f)$     | Effort factor: 0.5 / 1 / 1 for Easy / Hard / Max (2)                                            |
| $S_{g,k}$            | Effective weekly sets, muscle $g$ , week $k$ (3)                                                |
| $\varphi(w, r), K$   | Epley estimated one-rep max and its rep denominator ( $K=30$ default, self-calibrated) (4), (8) |
| $A_x$                | Anchor: best prior $\varphi$ for exercise $x$ (5)                                               |
| $L_{g,k}$            | Weekly peak $\varphi$ on primary exercises for $g$                                              |
| $D_g$                | Recent dose: recent mean of $S_g$ (11)                                                          |
| $\rho(s), s_0$       | Dose-response stimulus fraction and its scale (9)                                               |
| $T^S, T^L, \tau_g$   | Set trend, load trend, combined trend (12)–(13)                                                 |
| $\sigma_g$           | Per-muscle state (growing / holding / under-stimulated)                                         |
| MV, MEV, target, MAV | Landmarks 2, 6, 10, 20 effective sets/muscle/week                                               |

## B Parameter values

Every numeric constant in the model, its value, and where it enters. These are the values used in production (v121); the sensitivity analysis (Section 14) quantifies how much each one matters.

| Parameter                      | Value                                                                                          | Where                  |
|--------------------------------|------------------------------------------------------------------------------------------------|------------------------|
| Rolling window $N$             | 10 weeks                                                                                       | series length (3)      |
| Recent / earlier windows       | weeks 8–10 / weeks 5–7                                                                         | (10)                   |
| Secondary-muscle weight        | 0.5                                                                                            | $\alpha$ (1)           |
| Effort factors (Easy/Hard/Max) | 0.5 / 1 / 1                                                                                    | $\varepsilon$ (2)      |
| RIR $\rightarrow$ effort bands | Max $\leq 1$ , Easy $\geq 4$ , else Hard                                                       | (7)                    |
| Epley denominator $K$          | 30 default, fitted then clamped to [20, 40]                                                    | (4), (8)               |
| Dose–response scale $s_0$      | target/ $\ln 5 \approx 6.21$                                                                   | $\rho(s)$ (9)          |
| Maintenance volume MV          | 2 effective sets/wk                                                                            | state rule 1           |
| Min. effective volume MEV      | 6 effective sets/wk                                                                            | state rules 3–4        |
| Practical target               | 10 effective sets/wk                                                                           | radar goal; sets $s_0$ |
| Upper bound MAV                | 20 effective sets/wk                                                                           | plan forecast          |
| Sharp-drop trend cutoff        | $\tau_g < 0.75$                                                                                | state rule 2           |
| Growth trend cutoff            | $\tau_g \geq 1.08$                                                                             | state rule 3           |
| Maintenance-band climb         | $\tau_g \geq 1.15$ and $D_g \geq 5$                                                            | state rule 4           |
| Borderline margins             | $\pm 1$ set; trend within 0.06 of {0.75, 0.92, 1.08}                                           | Section 9.1            |
| Overall “at risk”              | $n_{\text{shrink}} > n_{\text{grow}}$ and $n_{\text{shrink}} \geq \lceil m/3 \rceil$           | Section 10             |
| Overall “growing”              | $n_{\text{grow}} \geq \max(1, \text{round}(0.4m))$ , $\tau_{\text{all}}$ absent or $\geq 0.95$ | Section 10             |
| “Need more data”               | fewer than 3 active weeks                                                                      | Section 10             |
| Plan frequency cap             | $\min(1.2, p/W)$                                                                               | (14)                   |

---

Yalla training-stimulus model, v121. Estimates a training stimulus from logs; not medical or diagnostic advice, and not a measurement of muscle mass.